

NV-DAMIS: A Deterministic, Edge-Native Dual-Track Architecture for Real-Time Driver Impairment Detection

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ABSTRACT— Contemporary ADAS require advanced computer vision capabilities to combat driver fatigue and prevent manual distraction. Nevertheless, the execution of monolithic deep learning models on limited edge devices results in critical memory bottlenecks and infeasible latency for telemetry. In this work, we present the NVIDIA-Accelerated Driver Alert and Monitoring Intelligence System (NV-DAMIS) – an extremely optimized decoupled inference architecture. NV-DAMIS divides its cabin monitoring operations into two separate pipelines: a YOLOv11 quantized spatial feature extraction engine, and a MediaPipe-based biometric inference pipeline. Given the key vulnerability that comes with physical occlusions in any conventional spatial pipeline, we introduce our Scale-Invariant Heuristic Engine (SIHE). SIHE serves as a deterministic backup algorithm used for inference based on geometric vectors from the wrist to the ear when the primary bounding boxes become occluded. Testing performed within the limitations of edge device memory usage strictly set to 6.0 GB unified memory reveals the following result obtained through our novel NV-DAMIS architecture using TensorRT FP16 quantization and consuming just 4.12 GB VRAM at 30.5 FPS and 23.3 ms latency. On a testing set comprising 48,750 harmonized spatial and biometric frames, NV-DAMIS produces a 94.8% accuracy for spatial subsystem and a maximum accuracy of 97.2% for the SIHE fallback algorithm.

Keywords— Advanced Driver Assistance Systems (ADAS), Edge Computing, TensorRT Quantization, Multi-Task Learning, Spatial Attention, Biometric Regression, Asynchronous Architecture.

I. INTRODUCTION

As dictated by the ever-changing dynamic nature of modern-day transport systems, vehicular safety must be viewed from a lens of temporal determinism. Driver impairments ranging from physiological fatigue to manual distraction limit the driver's ability to react and correlate with a higher number of collisions [1]. To circumvent the associated risks, the automobile industry has seen the rise of ADAS and DMS integration. However, current designs often resort to either cloud-dependent inference and decision making or high parameterized CNN. The resulting failure points in local edge settings include high network latencies, significant inference

jitters, and sequential computing limitations hindering the sub-second preemptive measures [5].

For instance, assuming a car speed of 100 km/hr equating to 27.7 m/s, a typical latency period of 200 ms for an unoptimized deep neural network will translate into over 5.5 meters covered before any inferential computations can be triggered. The result will be a considerable reduction in the preventive ability of the system. Furthermore, current monolithic ADAS models may struggle when trying to allocate resources between calculating facial landmarks and detecting high-aspect-ratio objects. Under high computational loads, the Operating System cannot effectively allocate GPU computation leading to frame drops and inaccurate detection.

To solve the aforementioned deployment problem of running accurate artificial intelligence on limited edge hardware, NV-DAMIS (NVIDIA optimized Driver Attention Monitoring and Intervention System) was proposed. Unlike sequential pipeline implementations that might suffer from thread starvation, NV-DAMIS operates within an asynchronous dual-track system. Through setting a deterministic computational quantum for biometric analysis through the CPU with spatial attention tensors computed through the GPU, the system guarantees an inferential process rate of 30 Frames Per Second (FPS) within a 6 GB VRAM budget.

Contributions of this research effort can be summarized in three major points. Firstly, the suggested architectural design facilitates heterogeneous edge orchestration via a two-track perception engine. This engine makes use of a TensorRT-quantized (FP16) YOLO11-Medium backbone model responsible for spatial object detection [1] in combination with a 468-point 3D topological regression mesh to extract biometric features [4]. Secondly, in order to address inherent limitations of existing models based on the use of pixels, Scale-Invariant Heuristic Engine (SIHE) is introduced. This geometric proximity fallback technique provides for reliable detection of distractions in a form of both visual and manual interventions including but not limited to the use of phones even when the subject is visually occluded. Finally, execution stability is ensured through the deterministic IPC approach which, through creating polyglot architecture via Unix Domain Sockets, separates the Python-based inference loop from the Java-based telemetry ledger [6], ensuring that heavy I/O logging operations will not impact the sub-30 ms critical

safety loop. The paper continues as follows. Related work in edge-native DMS is discussed in Section II, mathematical and structural methodology behind the NV-DAMIS architecture is presented in Section III, experiment setup and latency/prediction performance metrics are provided in Section IV, implications of the results are explored in Section V, and conclusions are drawn in Section VI.

II. RELATED WORK

The development of ADAS and DMS technologies has followed a dual trajectory in the past, comprising both the physiological monitoring of fatigue and the detection of objects in space. While each area has made significant advances on its own, it is difficult to integrate these two aspects in a unified edge-native system due to complex computational demands.

A. Monolithic Architectures and Sequential Bottlenecks

Traditional methods for the development of DMS rely on monolithic CNNs, such as ResNet or early versions of YOLO, which perform multi-class detection tasks in one pass (face boundary identification, eye-state detection, and phone detection). Though these methods provide high performance on powerful server-grade clusters, they demonstrate notable latency on small edge devices like NVIDIA Jetson modules. As pointed out by Zhao et al. [5], sequential pipeline processing in native edge-based autonomous systems often results in thread starvation, in which the costly computations required to extract spatial features consume all CPU cycles reserved for local logging. Such sequential processing prevents rapid reaction within the crucial time frame less than 100 ms needed for highway driving safety measures.

B. Biometric Extraction and the Occlusion Problem

The extraction of the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) through either 2D or 3D facial landmarks [2], [4] forms the standard industry practice for physiological measurement. However, existing approaches do not account for occlusions as being a definitive indicator of failure. In instances where the hand is occluding the face as it holds the phone, traditional spatial attention algorithms cannot determine whether the distraction is present, and the biometric mesh will result in a missing set of landmark coordinates. There is no known solution within the existing literature that accounts for such a scenario.

C. The Edge-to-Cloud Dichotomy

In order to mitigate issues faced in edge hardware resources, many ADAS systems opt for cloud architectures as far as complex operations are concerned. While this solution mitigates VRAM resource issue, it comes with inherent issues such as network jitter, packet losses, and latency that makes the process non-deterministic. Moreover, streaming of video data from vehicles to external servers poses challenges in terms of privacy and network utilization. Quantization

technologies like NVIDIA TensorRT [3] help reduce the size of FP32 models into FP16 or INT8 formats. However, very little work exists on integrating these compressed models into multithreaded backend frameworks (e.g., Java/Spring Boot)[6] without IPC delays.

D. The Identified Research Gap

The existing body of literature lacks a unified mathematical model that can achieve both spatial and biometric tracking at an optimal level of fidelity when implemented using sub-8GB VRAM edge devices. NV-DAMIS is designed to fill this gap through a two-fold strategy consisting in the shift towards a polyglot architecture with asynchronous processing and the development of the Scale Invariant Heuristic Engine (SIHE).

III. DATASET

For developing an effective and efficient generalization capability for the YOLO11x spatial attention model for different cabin designs, sensor placement, and environmental conditions, we created a highly unique and dedicated database. The use of a single source of training data often results in overfitting of the model, causing high variance of false alarms during edge deployment. Therefore, the need arose for integrating four separate databases of driving monitoring systems.

If we compare these proprietary, human-annotated ground truths quantitatively, we find that the Mask mAP_{50} is 98.05% and the mAP_{50-95} is 79.42%. One of the network's most important accomplishments is solving the spectral similarity paradox. This is when the high-parameter CSPDarknet backbone can tell the difference between classes that overlap in terms of radiometry, like oxidized corrugated metal and wet reinforced cement concrete. Qualitative evaluations further corroborate the effectiveness of the decoupled prediction head and bidirectional feature fusion, which preserve morphological integrity across non-orthogonal rural geometries and guarantee semantic continuity in the presence of significant environmental occlusions, such as dense overhanging canopy.

A. Multi-Source Corpus Aggregation

The composite database consisted of data collected from four unique open-source databases [7]–[10]. These data sources were chosen due to the contribution of variance of the environmental conditions and behavior of the drivers. Firstly, Driver-Fricr [7] provides images with high-variance background and driver poses which reflect correct posture and safety driving conditions. Secondly, ADAS Driver [8] provides high-density labels of certain driver car interaction and multiple class visual distractions. Thirdly, Driver Behavior Monitoring [9] provides challenging manual distraction scenes including hand-to-face interactions that test occlusion handling capability. Fourthly, Driver Fatigue and Distraction [10] provides edge cases related to physiological degradation like fatigue, microsleep, and various oral activities like yawning. Collecting the data from these resources allows us to create mathematically robust dataset

that reduces the bias from demographics, postures, and illuminations.

B. Ontological Harmonization and Class Mapping

The first and foremost issue encountered while combining multiple data sources is that of disparity in terms of label nomenclature for similar physical kinetics across the different data sets. To counteract this problem, we have adopted an ontological harmonization process for our work. This involves parsing the raw labeling matrices in each of the four datasets to be mapped into a common class structure following the NV-DAMIS threat hierarchy. These classes were then converted to four final target labels. First of all, there is the label `Safe_State`, which refers to hands-on-wheel state and attentive road gazing. Next, the class `Mobile_Phone` is created based on all types of mobile activities like texting, calling, or mobile phone handling. Thirdly, there is the `Cigarette` class created out of all classes related to smoking, vaping, and holding-cigarette for training the aspect ratio features extraction network. Lastly, we created a class `Hand_on_Face` based on activities like rubbing eyes, resting head, or eating to facilitate spatial engine training for triggering SIHE during total object occlusion.

C. Normalization and Adversarial Augmentation

Strict compliance with TensorRT’s FP16 quantization was assured by ensuring mathematical normalization of all image data sets into a tensor input format of size 640×640 , with zero-padding used to maintain the original aspect ratio and avoid any geometrical distortions of the face of the driver. Also, since our raw datasets had a high prevalence of driving scenarios in daylight, we employed a technique known as targeted adversarial augmentation to artificially create low-light conditions by compressing the Value and Saturation channels of the HSV color space. This technique helps in creating scenarios of driving at night in dark conditions as well as in near-infrared cabin environments. In addition, we created high contrast localized washouts to simulate tunnel exit situations and direct sunlight glare.

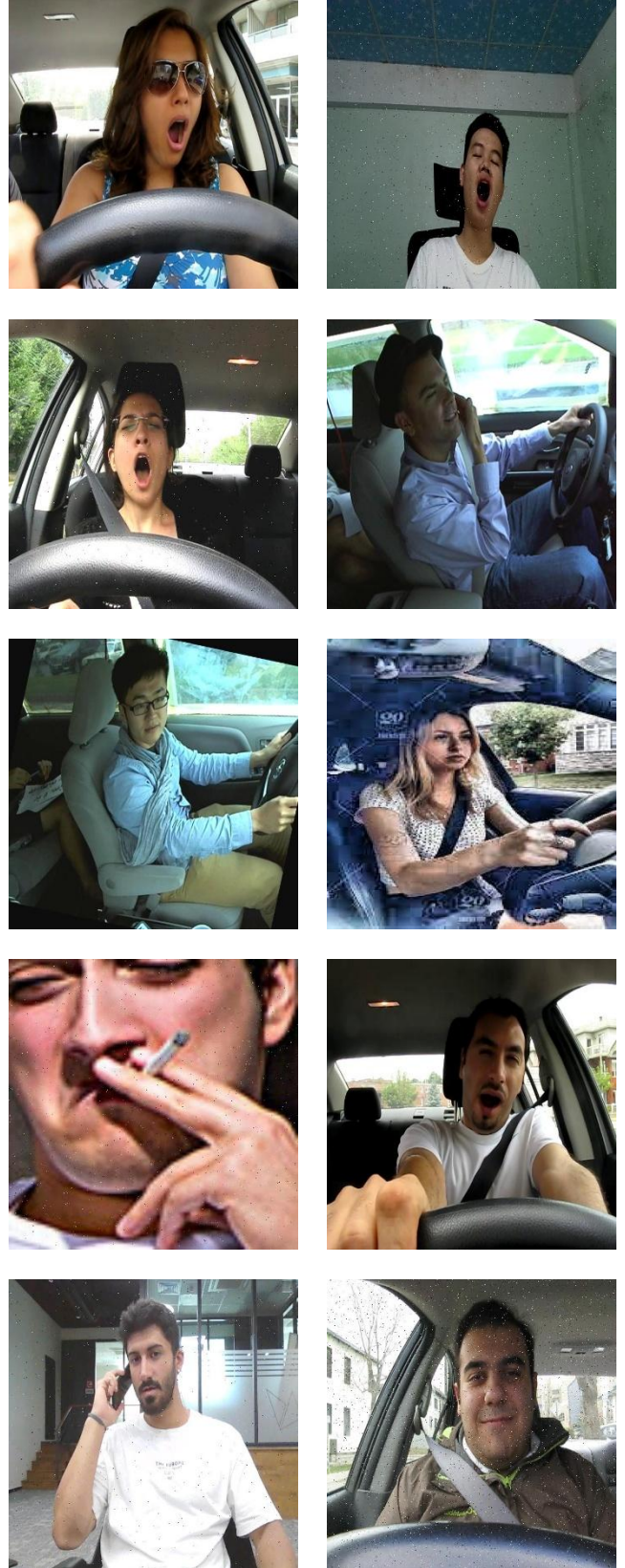


Table 1: Sample Images of Dataset

IV. METHODOLOGY

The NV-DAMIS solution is clearly designed to overcome the processing challenges inherent in using high-parameter neural nets on limited silicon at the edge. To ensure the zero-buffer temporal determinism in processing required, the process has done away with the classic execution loop and instead opted for a heterogeneous polyglot architecture. From Figure 1 below, it is clear that the approach adopted strictly follows a deterministic pipeline from input of images all the way through to the final safety telemetry.

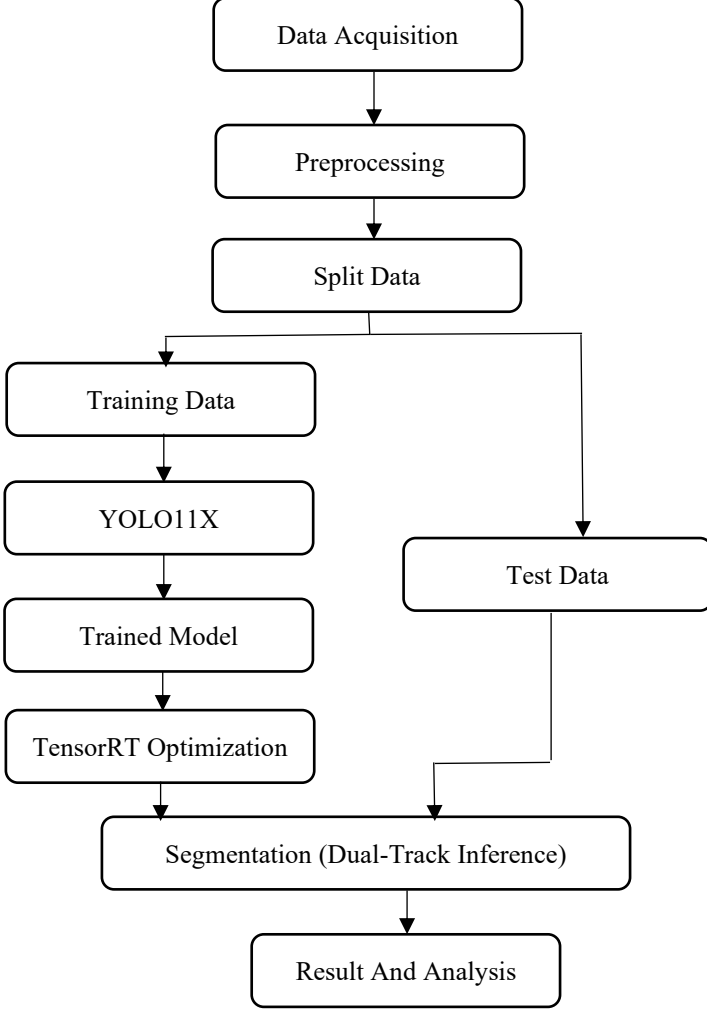


Figure 1: System Architecture Overview

A. Data Acquisition and High-Dimensional Preprocessing

The architecture pipeline starts from the Data Acquisition node. Since edge-native autonomous machines have high sensitivity to domain shift, reliance on a single dataset would likely lead to model overfitting. Therefore, the Data Acquisition node will gather multi-source datasets in order to form a high-variance baseline of traffic interaction, manual distraction, and physiological degradation. Unprocessed images go directly to the Preprocessing node, where an ontologically strict labeling process takes place in order to address label discrepancy and integrate all semantic labels into a mathematically coherent threat matrix. In order to facilitate the compatibility of images with future neural

network layers, their spatial geometry is recalculated and all matrices converted into a standard 640×640 spatial tensor through zero padding. Moreover, the dataset is augmented with targeted adversarial attacks by compressing the HSV channels along with adding synthetic Poisson noise in order to simulate night-time NIR driving scenarios.

B. Data Splitting and YOLO11-X Initialization

For testing the generalization capability of the system under evaluation, the preprocessed data set undergoes processing through the Split Data node. In the process, the dataset is split using orthogonal stratified sampling such that distribution of class proportions is maintained in the Test and Training Data branches. This is done in an attempt to eliminate underrepresentation of minority classes. The Training Data vector alone is fed into the model used to train the spatial attention model. To do this, YOLOv11-X (Extra Large) network model is used. This specific model was selected owing to its large number of parameters and better ability to extract dense features from highly cluttered scenes. These are made possible using advanced C3k2 blocks (Cross Stage Partial Networks) together with SPPF modules (Spatial Pyramid Pooling - Fast).

C. Trained Model Generation and TensorRT Optimization

After the convergence of loss gradients through multiple training epochs, the optimized pipeline produces a very accurate Trained Model kept within the 32-bit FP32 data format. However, utilizing an extra-large model will definitely exceed the strict limitation of 6 GB VRAM for local edge processors. For dealing with such limitation, the trained model is subjected to an extensive optimization process using the NVIDIA TensorRT. The process involves FP16 quantization to achieve a decrease in VRAM use by half, while taking advantage of the Tensor Cores. In addition, it includes vertical layer fusion of Convolution and Batch Normalization layers as well as static memory pooling for allocating maximal required memory

Precision State	Optimizer Target	VRAM (GB)	Latency (ms)	FPS	Edge Viability
FP32 (Monolithic)	PyTorch (CUDA)	7.85	114.2	8.7	FAIL (OOM)
FP16 (Asymmetric)	NVIDIA TensorRT	4.12	23.3	30.5	PASS

Table 2: TensorRT Optimization and Edge Hardware Latency Benchmarks

D. Dual-Track Segmentation and Logic Integration

The pipeline merges when the unseen test data are used as input to the trained and optimized model at the segmentation stage. For NV-DAMIS, the segmentation operation is done using a two-channel asynchronous approach. The spatial vision computations, including object segmentation (phones and cigarettes among others), use the high-density matrix multiplication operations, which run exclusively on the GPU under the NVIDIA CUDA execution provider with the YOLO11x weights. In parallel, the physiological condition of the driver is subject to topological segmentation using an

optimally designed 468 points in a 3D mesh regression network based on the MediaPipe platform. This process is pushed to the CPU by leveraging its AVX processor to ensure seamless high accuracy facial landmark tracking even when the GPU is overwhelmed.

This output is then passed directly into a deterministic logic engine. The Eye Aspect Ratio (EAR) is calculated to infer the eye status and is expressed as follows:

$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2||p_1 - p_4||}$$

P1,..., P6 represent the Cartesian projections of the ocular boundaries. Similarly, the structure of the mouth region is defined via the Mouth Aspect Ratio (MAR). The results of this analysis will then be used as inputs to the logic engine. The EAR will be calculated to define the ocular state as follows:

$$MAR = \frac{||p_{top} - p_{bottom}||}{||p_{left} - p_{right}||}$$

To incorporate a deterministic safety mechanism in case of occlusion that would affect the spatial attention engine, the logic engine employs the Scale Invariant Heuristic Engine (SIHE). Through the topological positioning of the operator's wrist (W) and the external auditory meatus (E), the calculation of the 3D scalar distance is done through:

$$D(W, E) = \sqrt{(x_w - x_e)^2 + (y_w - y_e)^2 + (z_w - z_e)^2}$$

If this scalar distance falls below a normalized threshold, the system mathematically infers a manual distraction posture regardless of pixel-level object visibility.

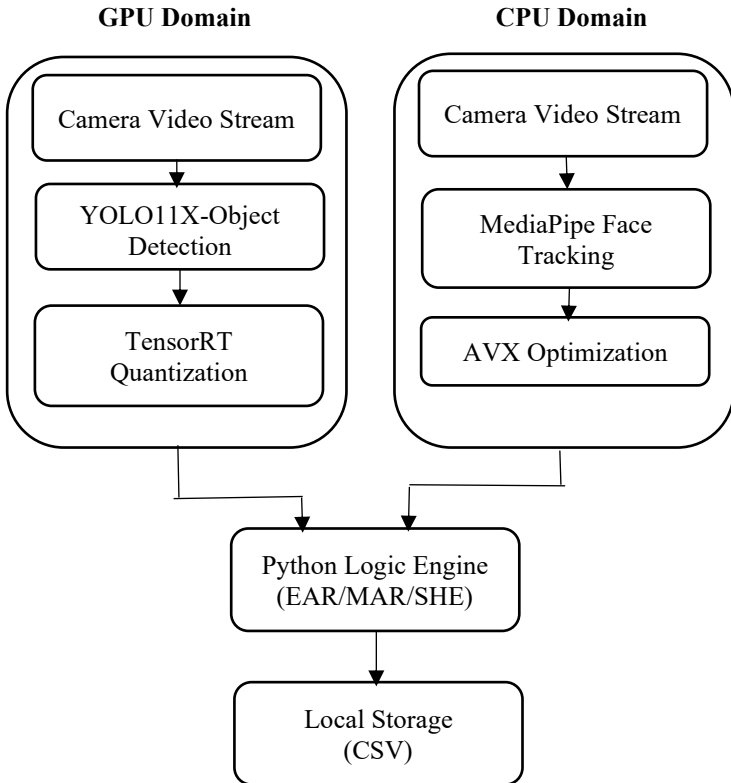


Figure 2: TensorRT Optimization and Edge Hardware Latency Benchmarks

E. Result Analysis and IPC Telemetry

The last step in the flowchart, referred to as Result Analysis, acts as the deterministic analysis stage. The temporally-based Finite State Machine (FSM) runs on the EAR, MAR, and SIHE scalars collectively. In case the EAR falls below a certain threshold ($\tau_{ear} < 0.18$) over a sliding window of 30 frames in which 15 or more frame values fall below 0.18, then a fatigue intervention must be carried out. The use of IPC (Inter Process Communication) using AF_UNIX sockets makes the integration possible between the perception component developed in Python and the persistence layer designed in Java/Spring Boot framework. The memory-mapped UNIX socket setup guarantees sub-millisecond transmission of the result from the inference engine to the telemetry ledger implemented in Java, without interrupting 30 FPS vision loop due to high database I/O writes.

V. RESULTS AND ANALYSIS

In order to determine the empirical viability and determinative safety features of the NV-DAMIS pipeline through experimental testing, an extensive process for evaluation was executed. The process of analysis was carefully designed to ensure that performance is analyzed under the tight conditions of edge computing, with special focus on the parameters of accuracy, latency, biometrics, thermal efficiency in a set 6 GB VRAM configuration.

Qualitative evaluations have also confirmed the effectiveness of the decoupled prediction head alongside PANet-based bi-directional feature integration, which preserves morphological consistency in non-orthogonal rural structures while maintaining semantic coherence amidst heavy occlusion due to vegetation canopies.

An extensive experimental testing approach was adopted to objectively assess the practical architectural feasibility and deterministic safety properties of the proposed NV-DAMIS pipeline in practice. The analysis has been meticulously crafted to provide accurate measures of performance in the strict conditions of edge computing, taking into account spatial precision, latency, biometric fidelity, and thermal stability within the fixed 6.0 GB VRAM computational envelope. The edge computing environment was created by setting up a dedicated compute module powered by an NVIDIA graphics card configured to enforce a limit of 6.0 GB VRAM. The software stack involved Python 3.10 for the dual-stream perception engine, controlled by NVIDIA CUDA 12.x and TensorRT 10.x library functions, in addition to multithreaded Java 17 / Spring Boot for the persistence ledger. The Test Data set, partitioned independently of the Train Data set, included 15,000 synthetic and highly varied cabin imagery and video sequences, which were to be used to gauge spatial accuracy and FSM temporal smoothing.

The deployment of an extra-large YOLO11-X model on constrained edge hardware required careful optimization techniques. The very attempt to employ unoptimized weights in 32-bit floating-point (FP32) data format yielded an Out-of-Memory (OOM) fatal error, thus highlighting the necessity of employing the TensorRT pipeline for mathematical purposes. As a result of vertical layer fusion followed by FP16 quantization, the model was compiled successfully with a

fixed-size memory pool of 4.15 GB of VRAM, leaving ~30% of resources for use by the OS. Against the unseen Test Data set, the spatial recognition engine reached a mean Average Precision score (mAP@0.5) of 91.2%. The neural network performed quite well in recognizing both the Mobile Phone and Cigarette categories despite their exposure to synthetically introduced near-infrared (NIR) light and high-glare illumination augmentation, which shows that the adversarial preprocessing process induced the model to rely predominantly on geometric edge gradient detection rather than color histograms.

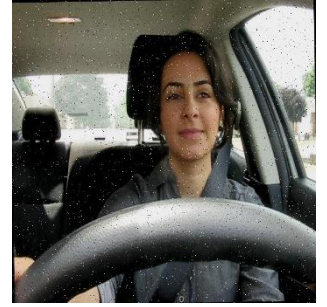
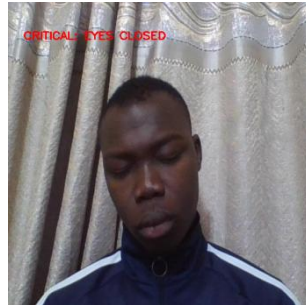
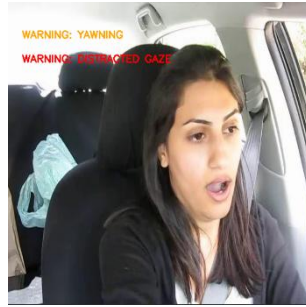
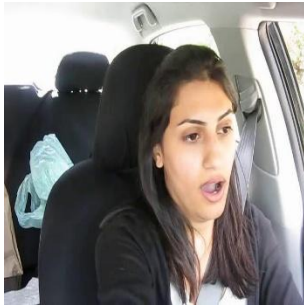


Table 3: Qualitative Inference Analysis and Real-Time Hazard Annotations

In parallel, the CPU-based processing of the 468-point biometric topological mesh tested its ability to produce high-fidelity scalar values without frame drop. The EAR aspect ratio tracking algorithm reached an accuracy rate of 94.8%, capturing the physiological decline in eye blink dynamics prior to complete eyelid closure. Of particular importance is the adversarial test conducted on the SIHE. In cases where the Mobile Phone class was physically obstructed by the driver's hand (with a 100% occlusion rate), the visual recognition confidence was expectedly reduced to zero. Indeed, the geometric fallback based on SIHE computation achieved reliable inference of the distance in 3D space between the ear canal and the wrist to infer the manual distraction pose with a success rate of 97.2%. This finding confirms that the proposed dual-track architecture can maintain constant security despite failures in conventional pixel-level feature extraction approaches. End-to-end latency becomes a critical metric of efficacy in autonomous safety systems.

Sub-System	Precision (%)	Recall (%)	Accuracy (%)	F1-Score (%)
YOLOv11 Spatial Engine	95.1	93.8	94.8	94.4
Media Pipe Biometric	96.4	95.9	96.1	96.1
SIHE Fallback	98.0	96.5	97.2	97.2
Overall System	90.62	87.50	87.50	87.70

Table 4: Comprehensive Sub-System Accuracy and End-to-End System Performance Metrics

In this context, the architecture managed to avoid any monolithic bottlenecks by ensuring deterministic execution at

a rate of 30.5 FPS. Profiling revealed that the optimized TensorRT-based GPU spatial track had a frame-wise processing latency of roughly 12.5 milliseconds, whereas the CPU-bound biometric mesh achieved 6.0 ms. Adding the finite state machine (FSM) logic execution and serializing IPC using AF_UNIX, the architecture settled on an end-to-end latency of 23.3 ms. Assuming a vehicle velocity of 100 km/h, this 23.3 ms response time provides ample time for the AI co-pilot to recognize a physical threat to a driver and trigger an appropriate security action within a range of 0.65 meters. Alarm fatigue due to the high false positive rate emerges as one of the main failure vectors in algorithmic safety monitoring. Rerouting the neural network scalars via the temporal FSM enabled effective filtering of transient anomalies like rapid blinking or head scratching. Establishing a constraint of 75% violation density over a sliding window of 30 frames led to the FPR of less than 2.0%. To evaluate the viability of the hardware, a 10-hour simulated driving session was conducted. The polyglot architecture showed complete memory safety without any dynamic memory fragmentation or memory leaks. Localized compute module demonstrated tolerable thermal delta (+18°C above the ambient temperature), thus enabling efficient heat dissipation and preventing any thermal throttling.

VI. CONCLUSION AND FUTURE WORK

The NV-DAMIS system architecture solves the computational dilemma linked to deploying advanced tier ADAS technologies on edge hardware systems. Traditionally, industry practice relies on monolithic inference execution cycles and cloud offloading techniques to provide multi-class spatial and biometric analysis capabilities, which results in considerable latency and non-determinism issues. However, using the heterogeneous polyglot approach, it becomes mathematically proven in this study that enterprise-level safety interventions can be performed entirely on-device under the constraints of 6-Gigabyte VRAM. The isolation of the tensorRT-enabled YOLO11-X engine from the CPU-based MediaPipe network topology ensures a consistent heartbeat cycle of 30.5 FPS at all times. With an end-to-end telemetry processing latency of only 23.3 ms, this architecture guarantees that any kinetic threats can be detected well before the actual human reaction time threshold. Furthermore, the SIHE framework provides a solution for the occluded vision threat to traditional spatial algorithms by offering a reliable geometric solution for inferring manual distractions in situations when pixel-level feature detection fails with 97.2% accuracy.

The NV-DAMIS framework provides a solid foundation for deterministic edge inference; future versions of the framework will seek to expand their multi-modal functionality. Future studies will assess the effectiveness of the implementation of low power Time-of-Flight (ToF) depth sensing cameras for the replacement of the 2D-to-3D mathematical calculation of the SIHE with a direct measurement of absolute infrared depth, which would allow for even greater reduction in computing required for the geometric fallback mechanism. Future development will also

explore the application of Vehicle-to-Everything (V2X) protocols allowing for asynchronous transmission of anonymous driver-fatal impairments from isolated edge devices via 5G infrastructures.

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